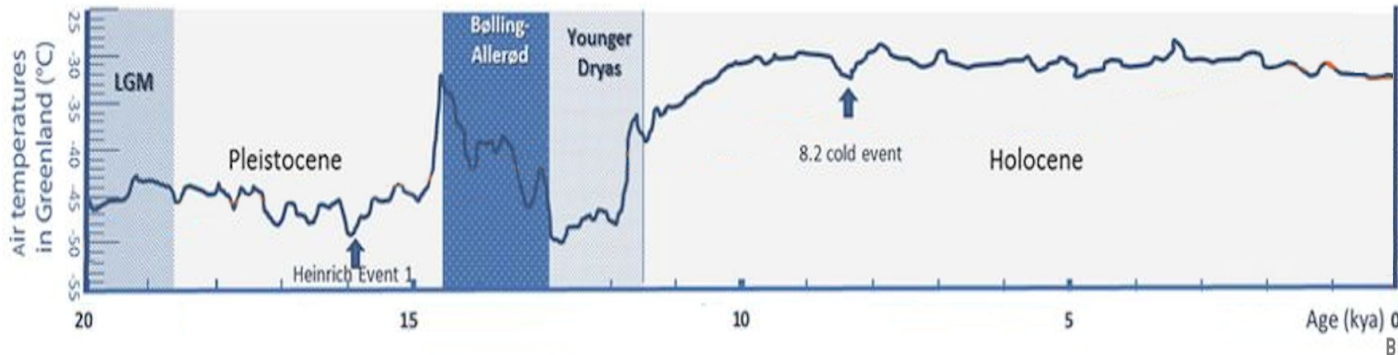
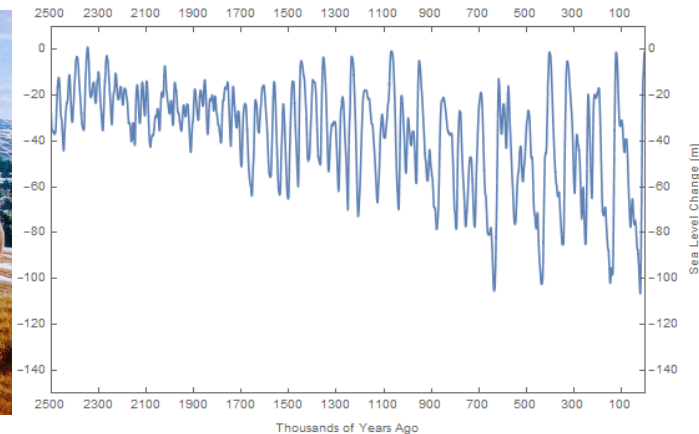
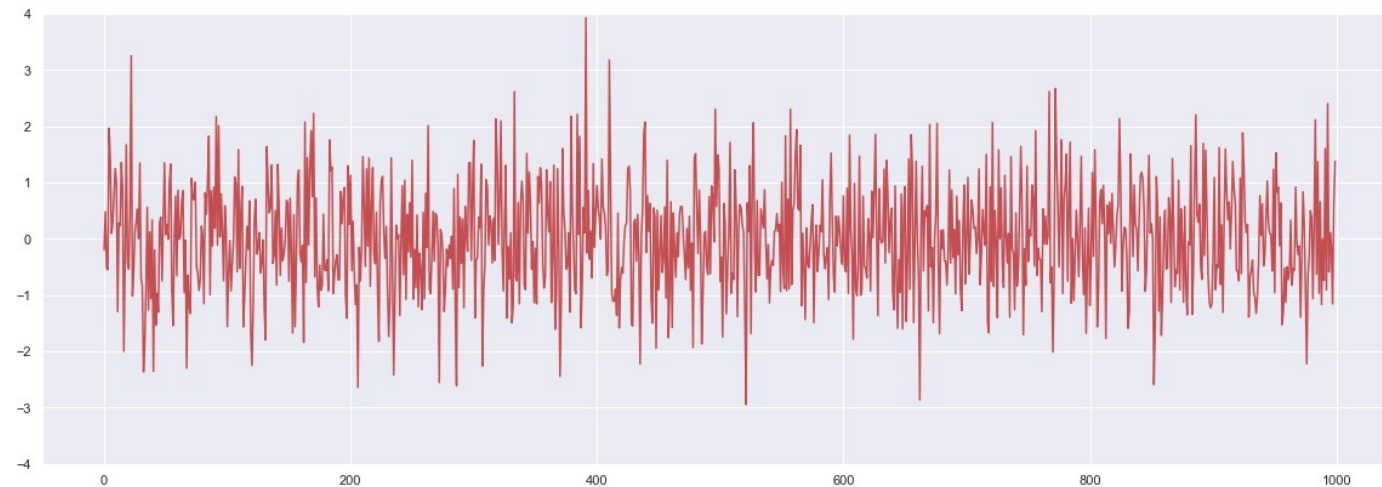


Evolutionary time series and selection signals

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia



Introduction to Time Series: White Noise and Random Walk



White Noise:

$$E[\epsilon(t)] = 0$$

$$\text{Cov}[\epsilon(t), \epsilon(t+k)] = \begin{cases} \gamma^2 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases}$$

Autocorrelation / temporal covariance

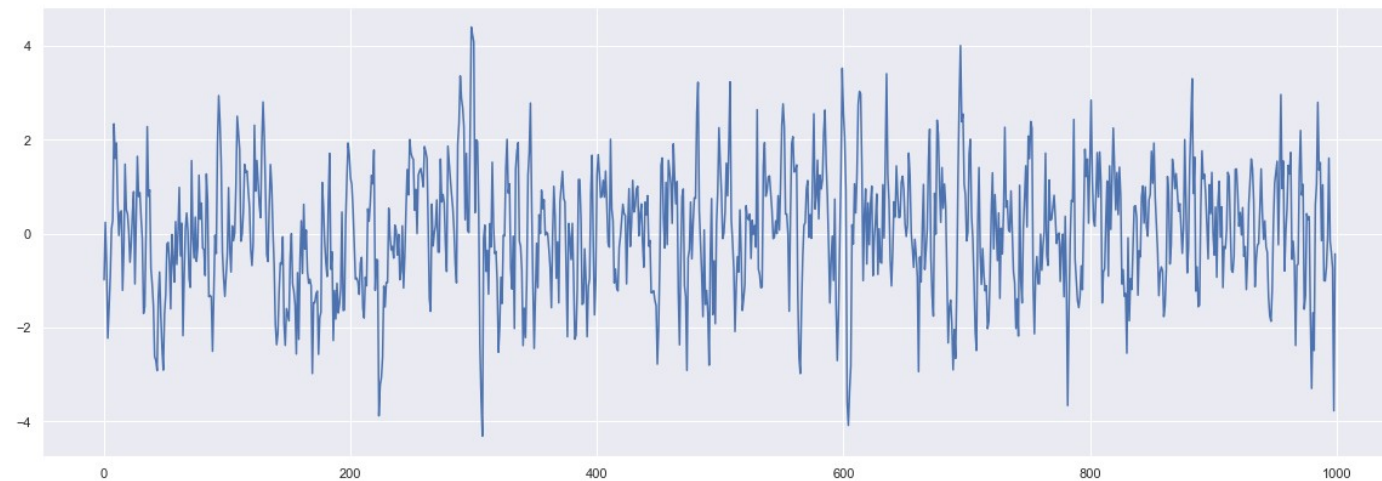
```
def covariance(x, lag):
    y1 = x[:len(x)-lag]
    y2 = x[lag:]
    covar = 0
    for i in range(0, len(y1)):
        covar = covar + (y1[i] - np.mean(y1)) * (y2[i] - np.mean(y2))
    return covar
```

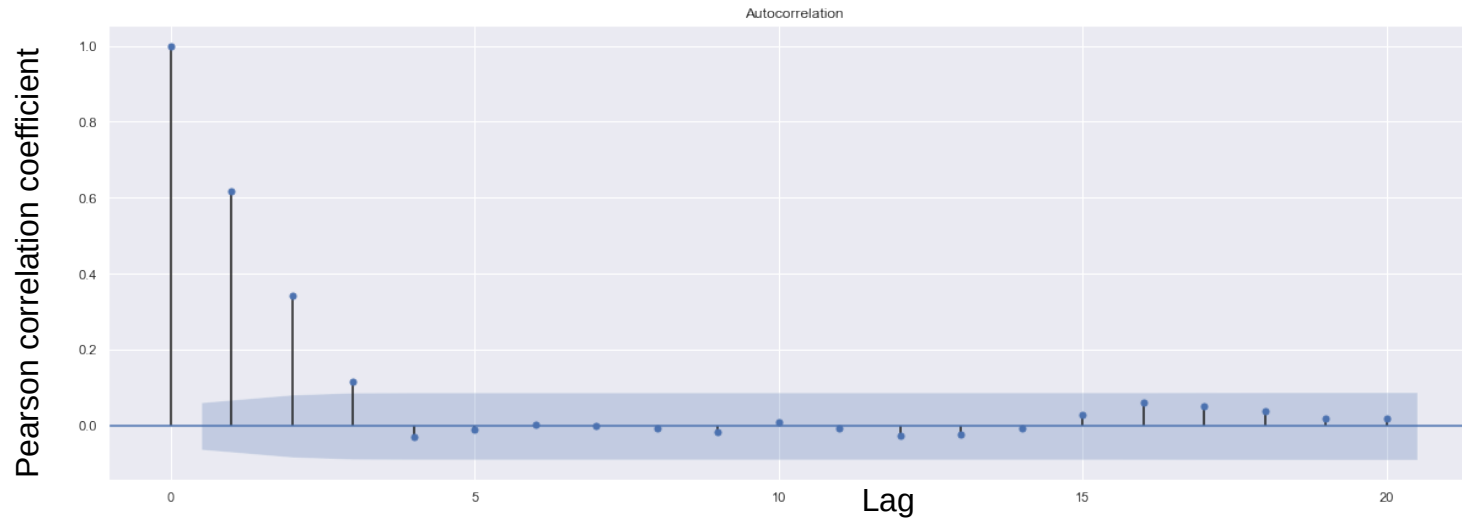
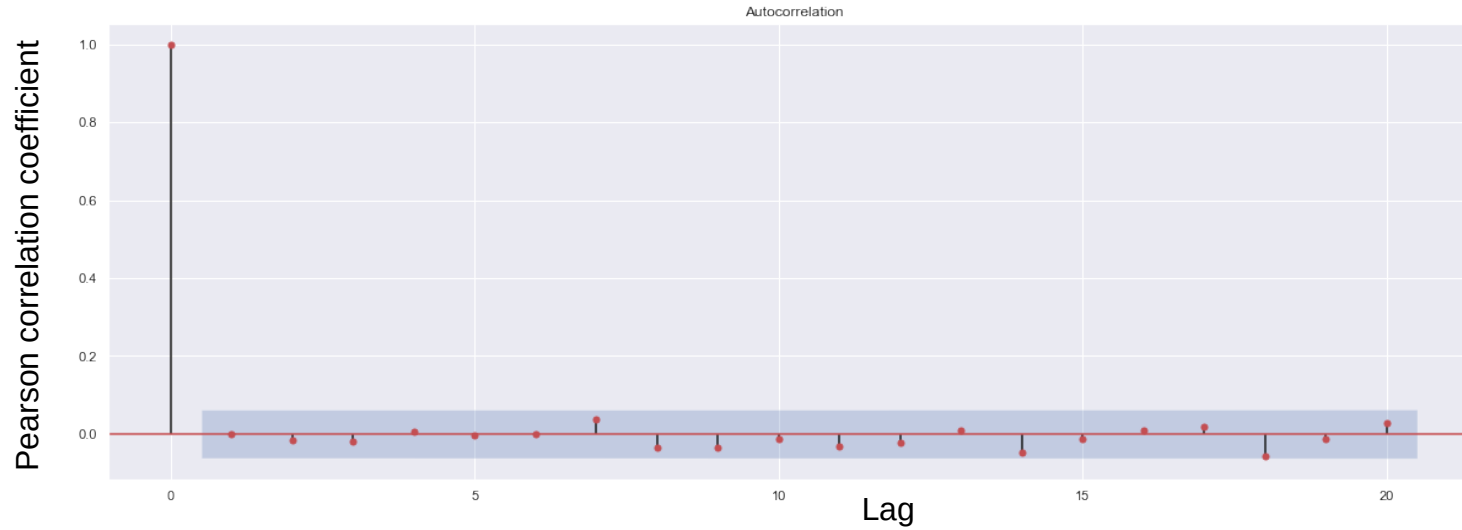
1, 2, 3, 4

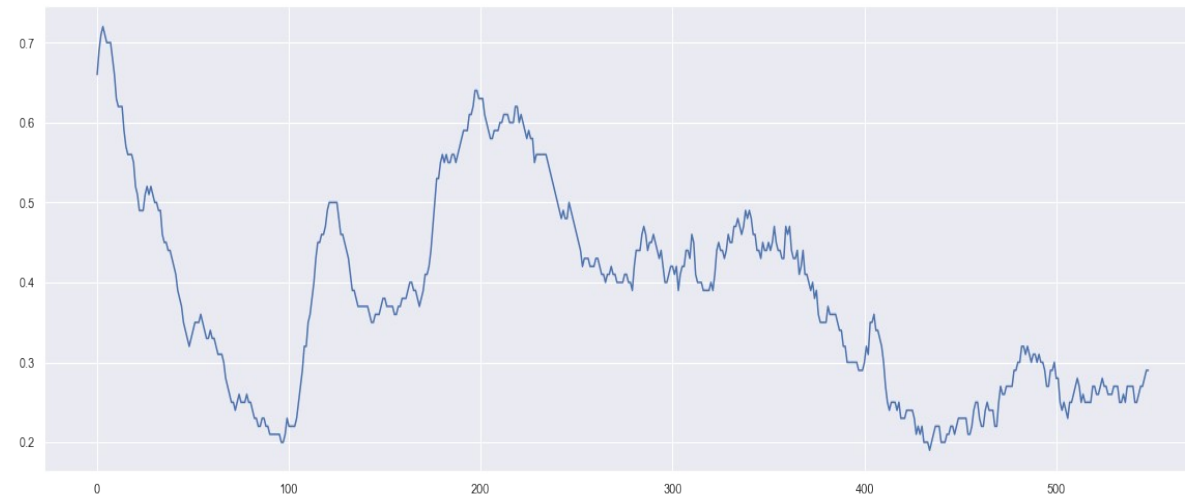
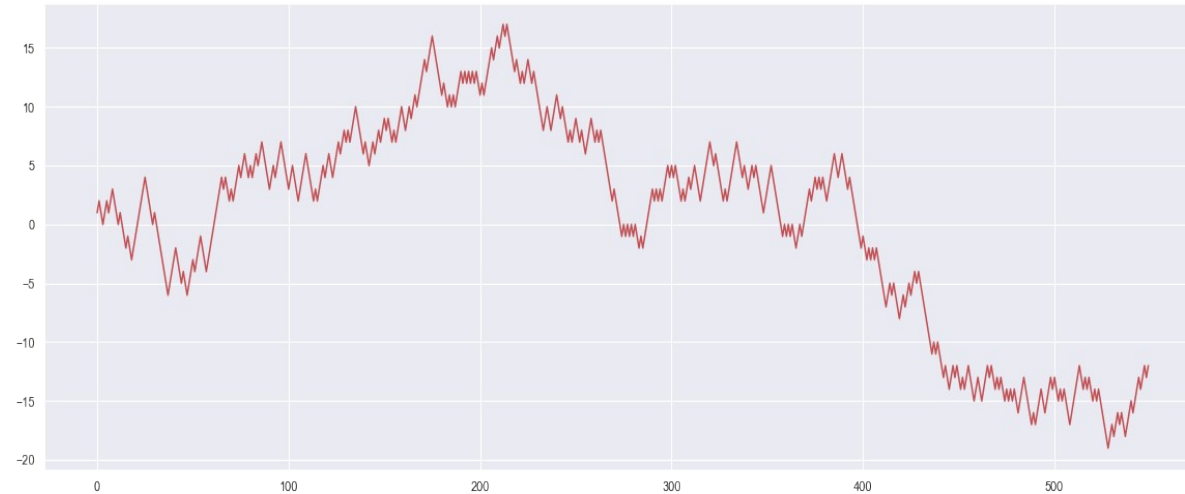
2, 2, 3, 4, 5



Pearson correlation





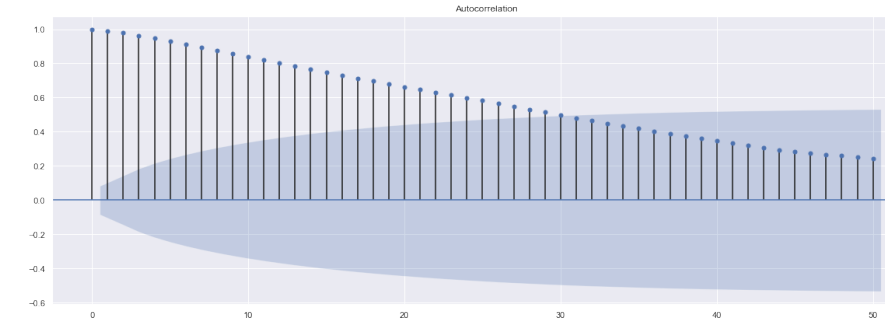
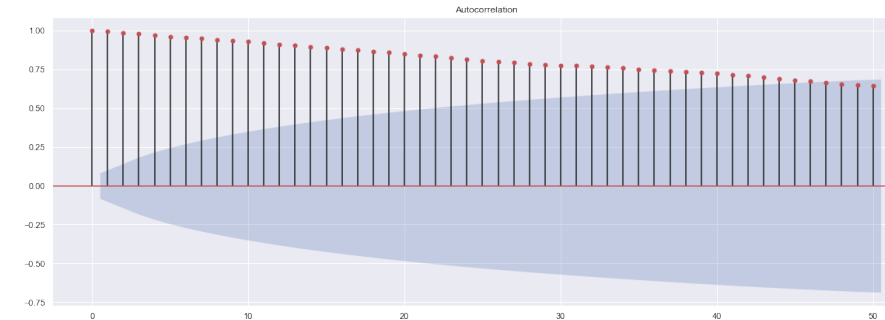


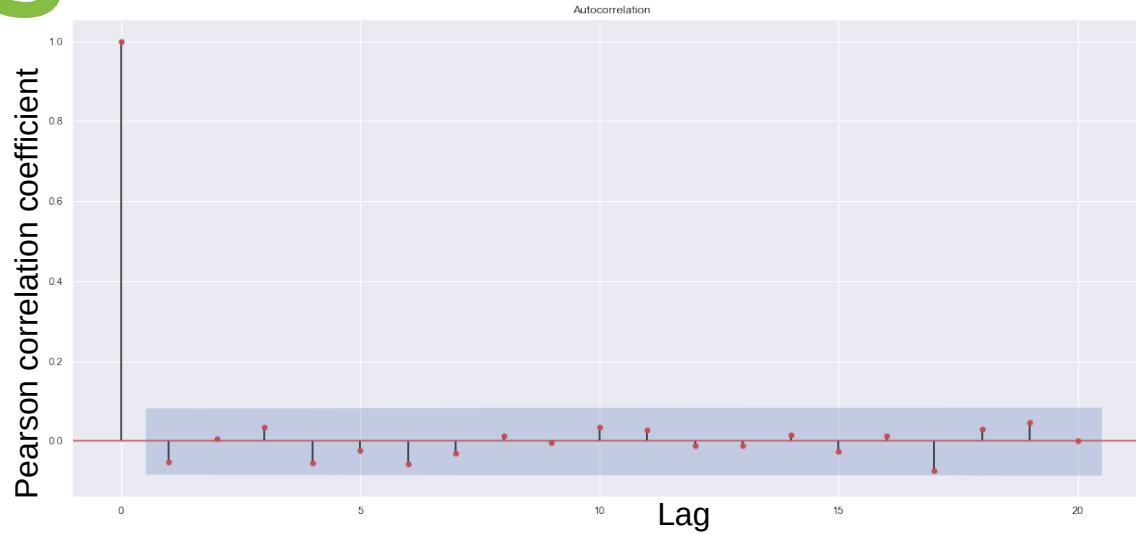
Markov chain:

White noise

$$X(t) = X(t - 1) + \epsilon(t)$$

$$\text{Cov}[X(t), X(t - k)] = \gamma^2(t - k)$$

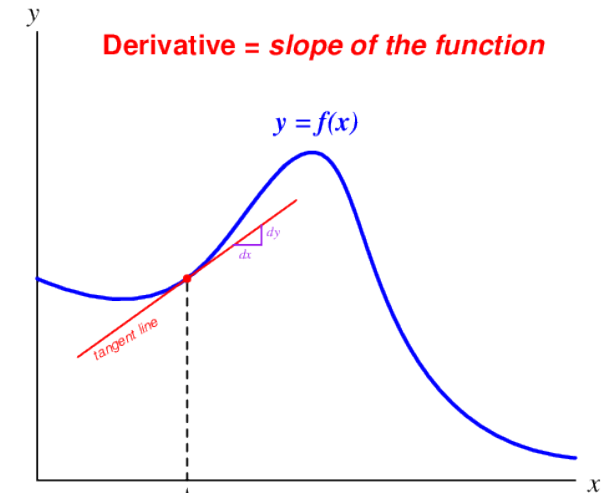
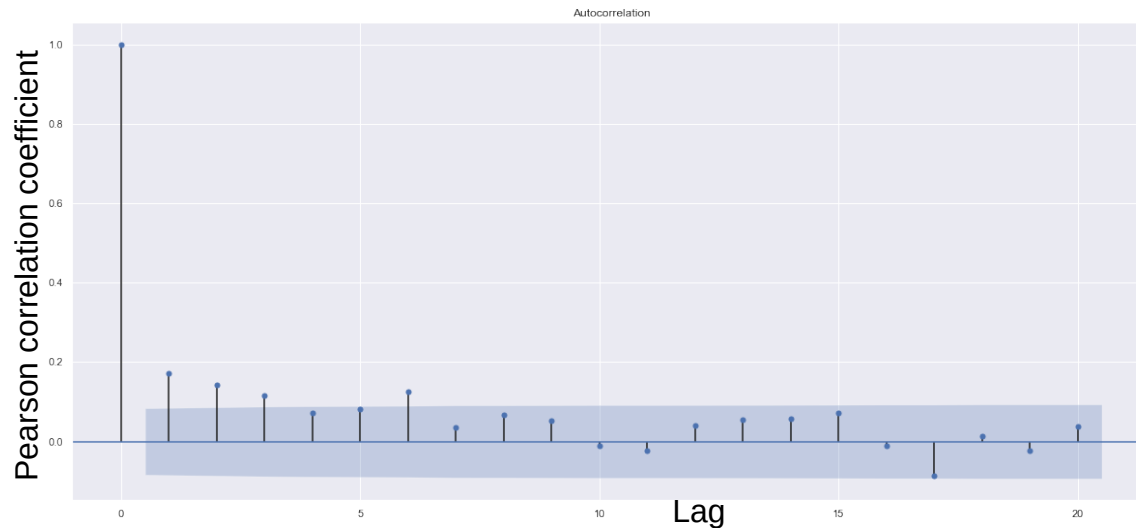




Derivative of a random walk is a white noise:

$$X(t) = X(t - 1) + \epsilon(t)$$

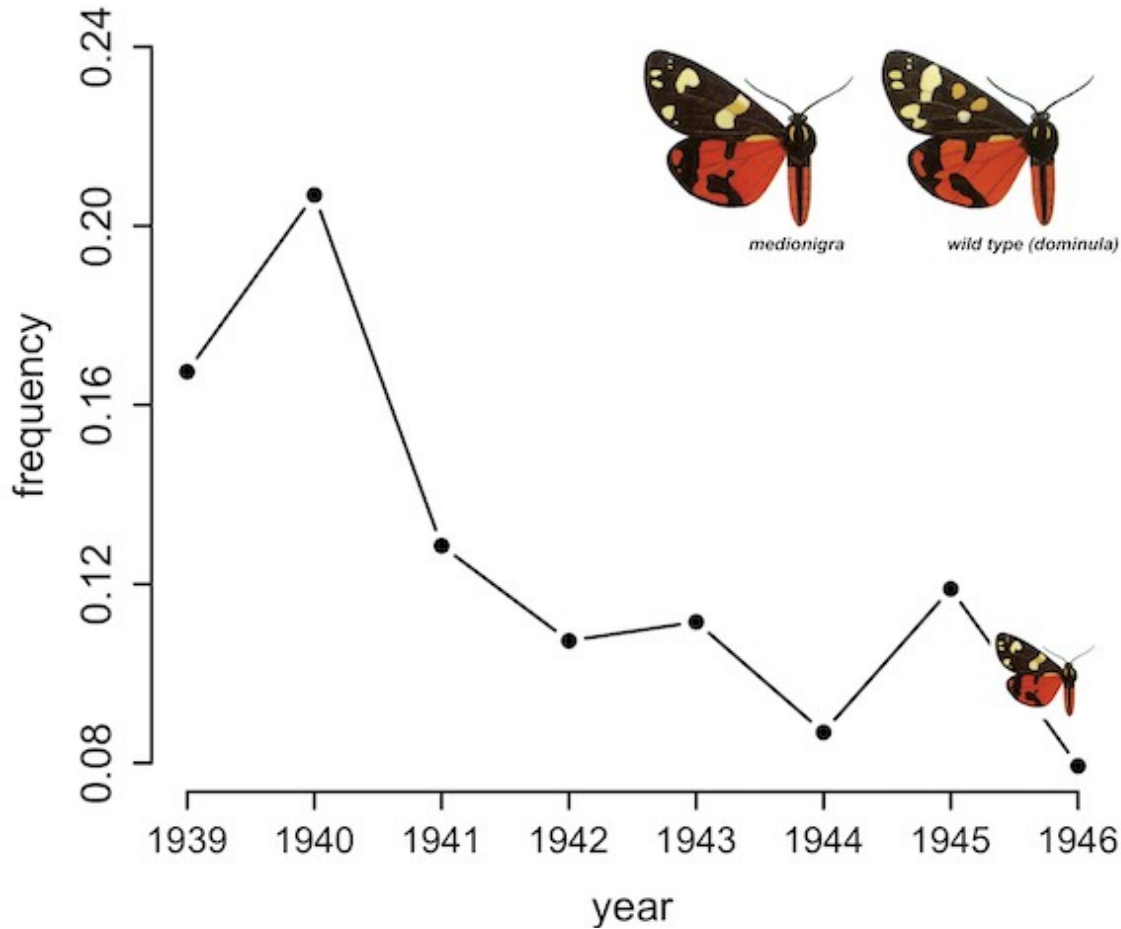
$$X'(t) = X(t) - X(t - 1) = \epsilon(t)$$



Slope at this point = $\frac{dy}{dx} = \frac{\text{Rise of tangent line}}{\text{Run of tangent line}}$

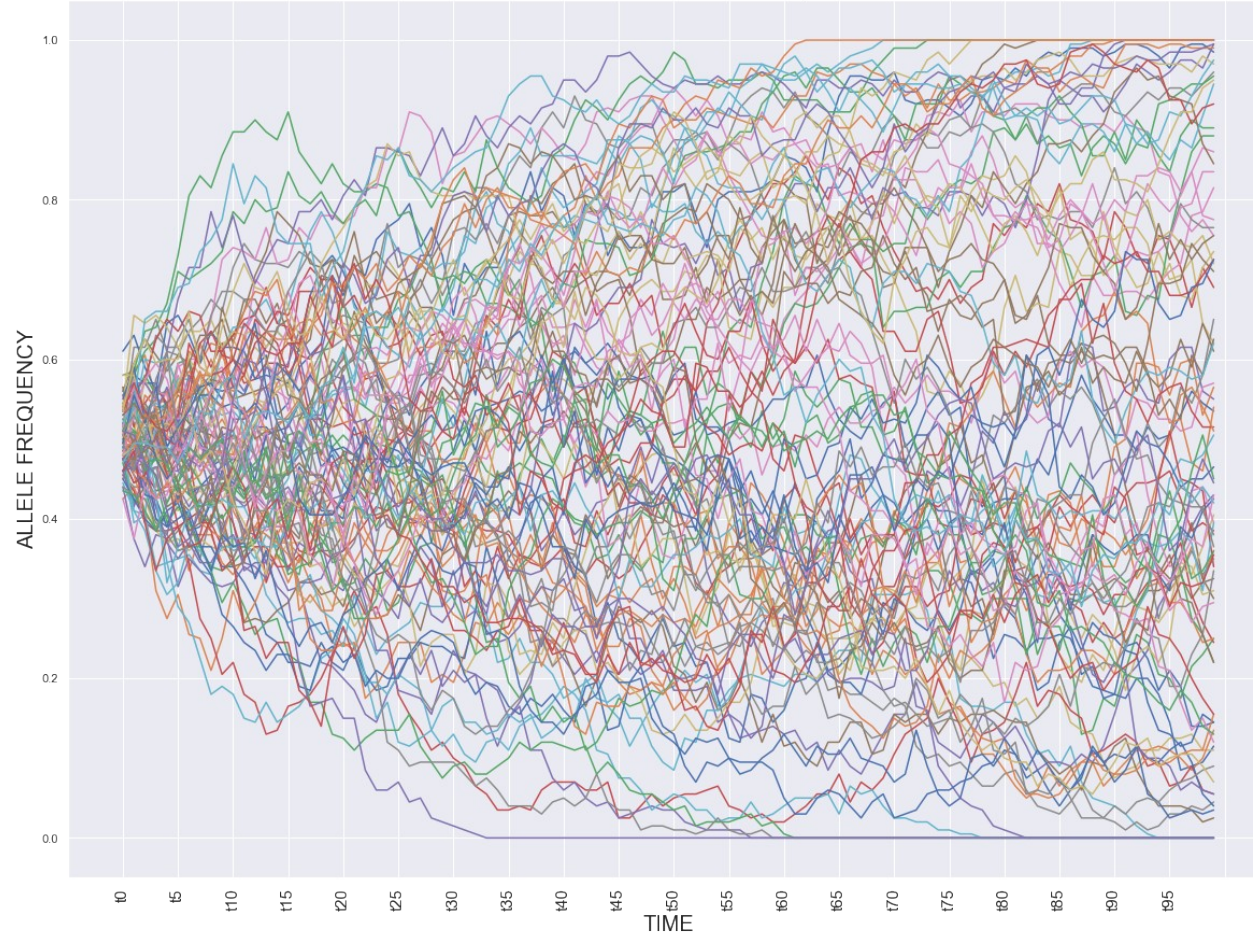
$\frac{dy}{dx} \approx \frac{\Delta y}{\Delta x} = \frac{\text{Small changes in } y}{\text{Small changes in } x}$

Allele Frequency Temporal Trajectory



Is the change in allele frequency purely random?

GENETIC DRIFT: WRIGHT-FISHER MODEL, POP SIZE N = 100



```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set()
np.random.seed(1)

# Allele counts follow binomial distribution: aka flipping a coin 500 times, tested 1000 times
N = 100          # population size
t = 100          # number of time points
freq0 = 0.5      # initial allele frequency
n_snp = 100      # number of SNPs

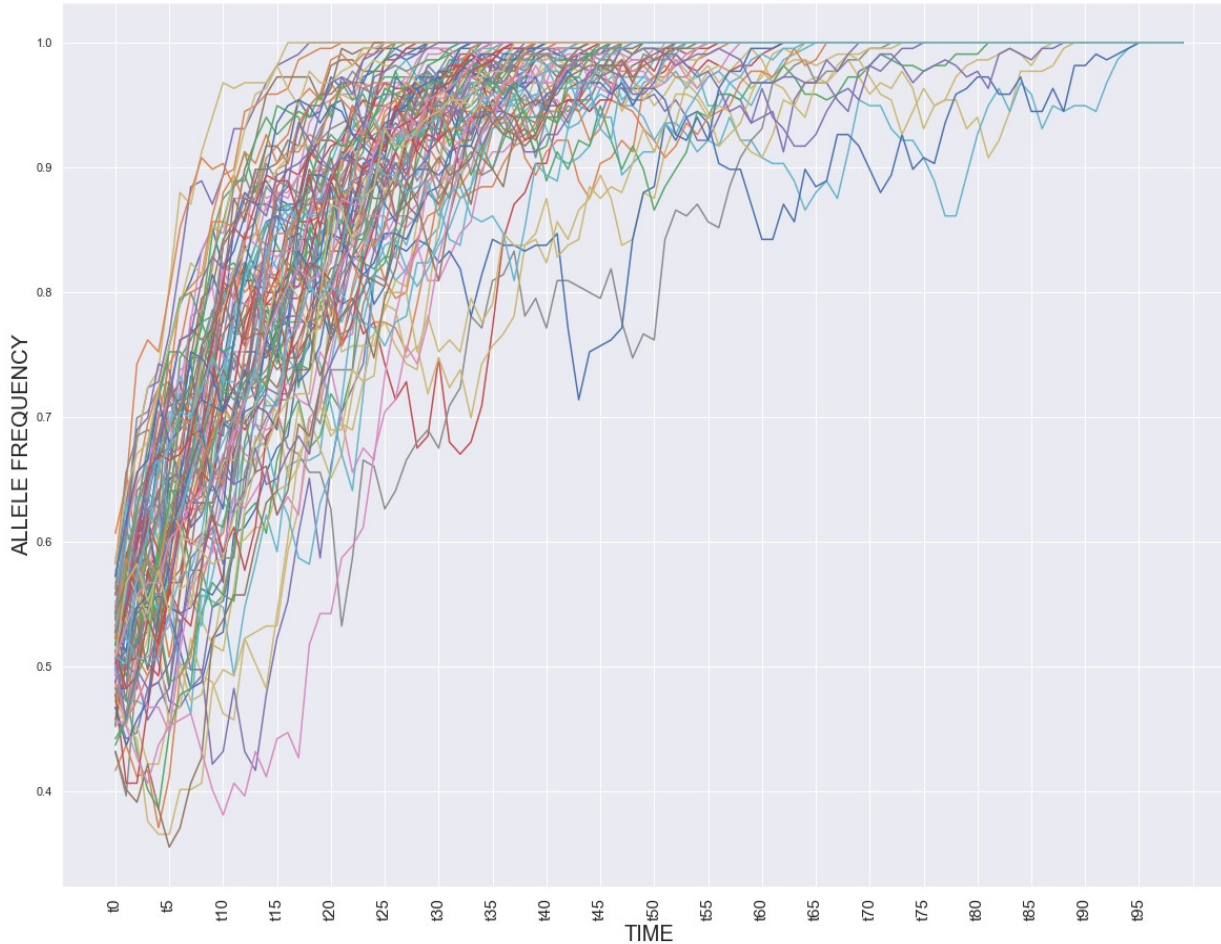
freq = np.empty(t + 1)
freq[0] = freq0
drift_df = pd.DataFrame(columns = [('t' + str(j)) for j in range(t)],
                        index = [('snp' + str(j)) for j in range(n_snp)])

for j in range(n_snp):
    for i in range(t):
        freq[i + 1] = np.random.binomial(2*N, freq[i], 1) / (2*N)
        drift_df.loc[snp + str(j), t + str(i)] = freq[i + 1]

drift_df.iloc[0:15, 0:20]
```

Genetic drift is a random walk
(Markov chain) from Wright-Fisher
model definition

WRIGHT-FISHER MODEL WITH SELECTION, POP SIZE N = 100



```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set()
#np.random.seed(12345)

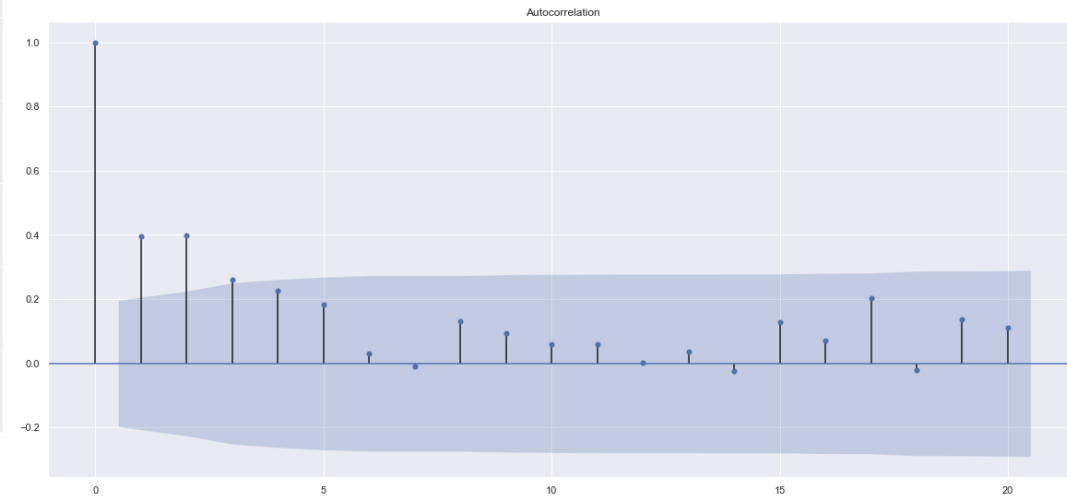
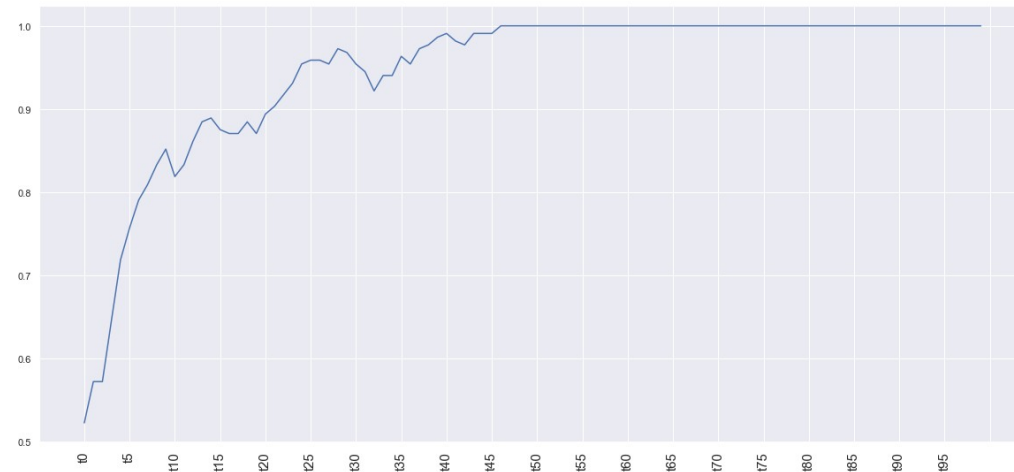
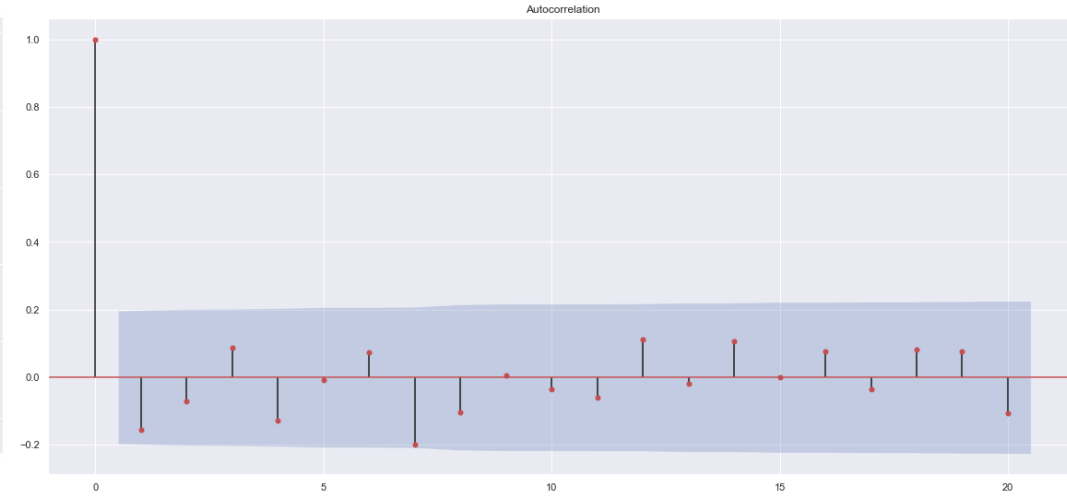
# Allele counts follow binomial distribution: aka flipping a coin 500 times, tested 1000 times
N = 100 # population size
t = 100 # number of time points
s = 0.095 # selection coefficient
freq0 = 0.5 # initial allele frequency
n_snp = 100 # number of SNPs

X = np.empty(t + 1)
freq = np.empty(t + 1)
freq[0] = freq0
sel_df = pd.DataFrame(columns = [('t' + str(j)) for j in range(t)],
                      index = [('snp' + str(j)) for j in range(n_snp)])

for j in range(n_snp):
    for i in range(t):
        X[i] = np.random.binomial(2*N, freq[i], 1)
        freq[i + 1] = X[i]*(1 + s) / (X[i]*(1 + s) + 2*N - X[i]) #X[i] copies of allele a and 2*N-X[i] copies of A
        sel_df.loc[ 'snp' + str(j), ('t' + str(i))] = freq[i + 1]

sel_df.iloc[0:15, 0:15]
```

Adding selection (s) makes allele frequency trajectory more directional (faster fixation)



Buffalo & Coop, PNAS 2020

Temporal Covariance Approach

Estimating the genome-wide contribution of selection to temporal allele frequency change

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Edited by Montgomery Slatkin, University of California, Berkeley, CA, and approved July 13, 2020 (received for review October 31, 2019)

Rapid phenotypic adaptation is often observed in natural populations and selection experiments. However, detecting the genome-wide impact of this selection is difficult since adaptation often proceeds from standing variation and selection on polygenic traits, both of which may leave faint genomic signals indistinguishable from a noisy background of genetic drift. One promising signal comes from the genome-wide covariance between allele frequency changes observable from temporal genomic data (e.g., evolve-and-resequence studies). These temporal covariances reflect how heritable fitness variation in the population leads changes in allele frequencies at one time point to be predictive of the changes at later time points, as alleles are indirectly selected due to remaining associations with selected alleles. Since genetic drift does not lead to temporal covariance, we can use these covariances to estimate what fraction of the variation in allele frequency change through time is driven by linked selection. Here, we reanalyze three selection experiments to quantify the effects of linked selection over short timescales using covariance among time points and across replicates. We estimate that at least 17 to 37% of allele frequency change is driven by selection in these experiments. Against this background of positive genome-wide temporal covariances, we also identify signals of negative temporal covariance corresponding to reversals in the

a polygenic trait (such as fitness) is distributed across numerous loci. This can lead to subtle allele frequency shifts on standing variation that are difficult to distinguish from background levels of genetic drift and sampling variance. Increasingly, genomic experimental evolution studies with multiple time points, and in some cases multiple replicate populations, are being used to detect large-effect selected loci (30, 31) and differentiate modes of selection (32–34). In addition, these temporal-genomic studies have begun in wild populations, some with the goal of finding variants that exhibit frequency changes consistent with fluctuating selection (35, 36). In a previous paper, we proposed that one useful signal for understanding the genome-wide impact of polygenic linked selection detectable from temporal genomic data is the temporal autocovariance (i.e., covariance between two time points) of allele frequency changes (37). These covariances are created when the loci that underlie heritable fitness variation perturb the frequencies of linked alleles; in contrast, when genetic drift acts alone in a closed population, these covariances are expected to be zero for neutral alleles. Mathematically, temporal covariances are useful because it is natural to decompose the total variance in allele frequency change across a time interval into the variances and covariances in allele frequency change between generations. Further-



vince buffalo

The Problem of Detecting Polygenic Selection from Temporal Data

The last chapter of my dissertation with Graham Coop was recently published in PNAS (pdf, bioRxiv) last week. In an effort to communicate my research to a broader audience, I have written two blog posts on our work. The first post, below, is meant to introduce the historical context and concepts like linked selection and polygenic adaptation to a non-scientist, and the second post describes our work on temporal covariance as a signature of polygenic linked selection and its application to four evolve-and-resequence data sets.

Natural Selection is Rapid

For nearly seventy years, we have known Charles Darwin was wrong about a key aspect of natural selection: that it was slow acting. In the first edition of *The Origin of Species*, Darwin wrote of natural selection,

We see nothing of these slow changes in progress, until the hand of time has marked the long lapse of ages, and then so imperfect is our view into long past geological ages, that we only see that the forms of life are now different from what they formerly were.

While Darwin knew artificial selection¹ could rapidly change an organism's traits, it was only until ecological geneticists E.B. Ford and Bernard Kettlewell showed rapid changes in moth wing coloration, that it was recognized that natural selection could act over very short timescales. Since, evolutionary biologists have shown rapid adaptation is remarkably common in a variety of species including sticklebacks², *Anolis* lizards³, soapberry bugs⁴, guppies⁵, and field mustard⁶.

Natural Selection versus Drift

Barghi et al. (2019) Data Processing and Analysis

Setup

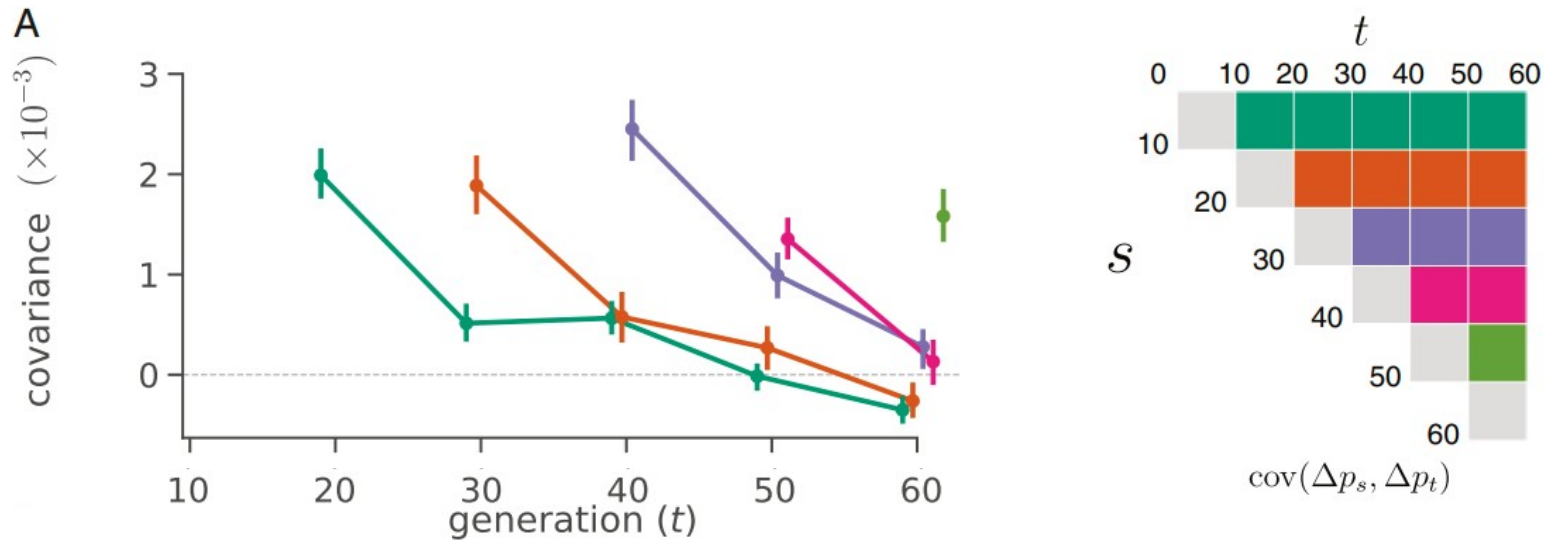
```
In [1]: import os
import sys
nb_dir = os.path.split(os.getcwd())[0]
if nb_dir not in sys.path:
    sys.path.append(nb_dir)
%load_ext autoreload
%autoreload 2
%load_ext autoreload

In [2]: import re
import pickle

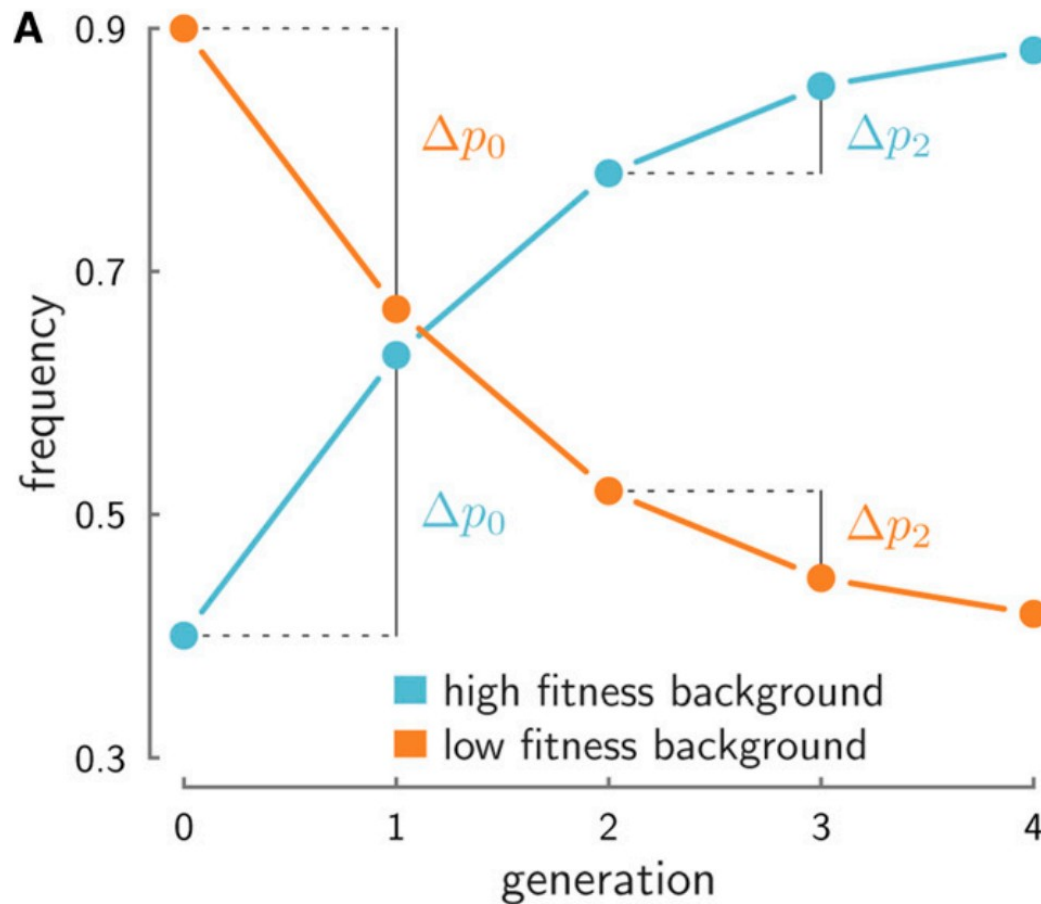
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib import cm
import statsmodels.api as sm
import matplotlib as mpl

In [3]: from cvtk.cvt import TemporalFreqs, TiledTemporalFreqs
from cvtk.cov import stack_temporal_covariances
import cvtk.variant_files as vf
from cvtk.gintervals import GenomicIntervals
from cvtk.pca import FreqPCA
from cvtk.plots import rep_plot_pca, correction_diagnostic_plot
from cvtk.utils import integerize
from cvtk.utils import extract_empirical_nulls_diagonals, extract_temporal_cov_diagonals
from cvtk.cov import stack_replicate_covariances, stack_temporal_covs_by_group
```

Notebooks are poorly commented,
hard to run and understand
(personally for me)



- What is a temporal covariance?
- Why do you compute the temporal covariance on Δ allele frequency?
- Why do you expect a temporal covariance to be positive under selection?
- Why can't genetic drift demonstrate a non-zero temporal covariance?



If we were to track this neutral allele's frequency trajectory through time in the entire population, we would see it rise between the two first consecutive generations, and then rise again between the next two, as long as it remains associated with a fraction of the father's beneficial genes. The neutral allele's frequencies between the second and first generation (mathematically, $\Delta p_1 = p_2 - p_1$), and the third and second generation ($\Delta p_2 = p_3 - p_2$) would both change in the same direction when they are associated with a good background, creating positive temporal covariance in the frequency trajectory.

In contrast, genetic drift cannot create positive temporal covariance in neutral allele frequency trajectories. Another way to say this is that when different chromosomes have heritable fitness differences, the frequency change of an allele at one time interval can be predictive of the changes at later generations, as long as the allele is still associated with its fitness background, and that background has the same effect on fitness.



*Knut och Alice
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